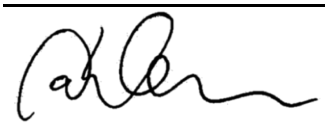
UNIVERSITY OF CAPE TOWN

Department of Electrical Engineering

EEE4022F/S - Final Year Project

Graduate Attribute Tracking Form

Student name: Thomas Kruger		DP Awarded? [Y/N]	Y
Student no:	KRGTHO002	Supervisor name:	A. Khan
Date:	16/09/2026	Date:	25/09/26
Student signature:		Supervisor signature:	

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Instructions:

Students must explain in this document what they **have already done** and what they **plan to do** to satisfy each Graduate Attribute. Descriptions of each GA is provided at the end of the document. Supervisors respond to the student's plans and current progress, providing additional comments or advice as they see fit. Once the student's progress is deemed sufficient (a few weeks before submission at the due date for this form), supervisors indicate that DP can be awarded.

VERY IMPORTANT: Receiving DP for the course does NOT imply that all GA's have been met in the course. Assessment of GA's only happen in the final marking of the project report.

GA 1: Problem Solving

Student Response:

My project concerns the field of LiDAR for wind-speed measurement. Specifically, my project deals with the problem of designing a prototype benchtop-scale version of a Doppler wind LiDAR system. I consulted various sources of literature in order to first understand what the key use-cases and goals of Doppler wind LiDAR systems are (mostly with respect to wind-turbines, and the use of such systems to control the turbines optimally). Thereafter, with the end-use in mind, I reviewed sources which discuss common types/classes of such systems, and identified a system type which would be suitable for this project, given the time-frame and budget – this type being coherent detection Doppler LiDAR.

After identifying this suitable class of system, I reviewed the various design considerations of such a system in the available literature. This included mathematical and physical descriptions of how the system operates, with respect to signal theory, optics, electronics, and relevant physics phenomena (including Rayleigh and Mie scattering, coherent detection methodology, balanced detection, and laser source physics). From this, I synthesised a list of parts which would be suitable for use in the

project, and used the relevant mathematics to validate these choices in terms of the final requirements.

Still to be done is the physical build of the system, and troubleshooting of its setup. Thereafter I will conduct testing in order to validate the design for the end use-case, and I will make recommendations where appropriate to improve the system.

Supervisor Response:
Student's problem-solving ability adequately demonstrated through approach to tackling research questions associated with final year project. In particular, student demonstrated ability to find and understand relevant literature. Thereafter, the student used appropriate theoretical formulations related to the topic to design a LiDAR system for wind speed measurement. The student is therefore on track meet requirements for GA-1.

GA 4: Investigations, Experiments, and Data Analysis

Student Response:
I conducted a thorough investigation into the various types of Doppler wind LiDAR systems which are used in industry and scientific research. After doing so, I conducted more specific research into coherent detection Doppler LiDAR systems, and the physics, signal processing and optics which underpin these systems. From this, I was able to recommend to my supervisor a course of action for the class of system to be designed, and substantiate this choice through the research done.

While not yet completed, I have formulated various experiments to conduct on the system once it has been built, which will validate the signal theory, mathematics and electronics principles concerned. The primary considerations in these experiments is a thorough noise analysis, which will serve to quantify the sensitivity of the system with respect to the atmospheric conditions in which wind-speed can be measured. Furthermore, the effects of various design decisions on the performance (e.g. source laser linewidth and power, or the use of balanced detection over single-ended detection) will be analysed in these experiments, and optimal requirements for such systems will thus be experimentally shown.

Supervisor Response:
Student demonstrated a sound investigative approach by conducting an extensive review of literature on Doppler wind LiDAR systems and a detailed assessment of system architectures. The proposed experimental work is appropriate and includes consideration of key performance metrics, noise analysis, and validation of theoretical predictions. The student showed ability to formulate investigations that will generate meaningful data and support evidence-based conclusions. The student is therefore on track to meet the requirements for GA-4.

GA 5: Use of Engineering Tools

Student Response:
I am currently in the process of setting up simulations using the Python language, and various publically available packages (such as OptiCommPy, NumPy, SciPy). These simulations will serve to validate and quantify performance of the system from the design which was formulated, with respect to noise, sensitivity, and component characteristics.

I will also be using the Python language for data analysis after the system has been built, and experiments run. An oscilloscope will be used for data collection, owing to its fast sampling rate, and the data will be exported to a laptop. This data can then be transformed in the relevant manner using Python (e.g. Fourier transform from time-series data, averaging of Fourier spectra) in order to present meaningful results for the system's capabilities.

Supervisor Response:
The student demonstrated competence in selecting appropriate engineering and computational tools for modelling, simulation and experimental evaluation of their system. The planned use of Python and publically available photonics simulation packages to predict system performance, together with oscilloscope-based data acquisition and subsequent signal processing, demonstrates appropriate application of engineering tools to the project. The student is therefore on track to meet requirements for GA-5.

GA 6: Professional and Technical Communication

Student Response:
I have attended scheduled meetings (and scheduled further follow-up meetings) with my supervisor in order to discuss the route which the project should take, as well as budget and technological considerations. I have also consulted with Maysam Soltanian to obtain suitable sources for components which are required for the project.

I have also completed a literature review which communicates clearly what the end-use cases of Doppler wind LiDAR systems are, and what the different classes of such systems are. The review also serves to explain physics, signals and mathematical theory which underpins the workings of coherent detection Doppler wind LiDAR systems.

Supervisor Response:
The student communicated project progress effectively through regular engagement with the supervisor and relevant technical staff. The literature review demonstrate their ability to present background research, underlying theory and project objectives in a structured and technically appropriate manner. The student is therefore on track to meet the requirements for GA-6.

GA 8: Individual Working

Student Response:
I have maintained goals and suggested timeframes for my workings on the project. While meetings with my supervisor and Maysam helped guide the sourcing and budget constraints for the project components, all of the specific components were selected by me, and suitable sourcing suggestions were made to Maysam. While the supervisor provided the overarching topic of Doppler wind LiDAR, I used my own research into the area to select an appropriate subclass of LiDAR system to build for the project, synthesising the constraints of the project and the desired requirements. All simulation and experimental working was synthesised individually.

Supervisor Response:
The student took initiative to research the field independently and selected a suitable system architecture, specifying components and planning the simulation and experimental work. While guidance was provided on the overall project direction, the technical decisions and implementation approach was largely developed by the student. The student is therefore on track to meet the requirements for GA-8.

GA 9: Independent Learning Ability

Student Response:
I have independently learnt concepts in photonics (laser source characteristics), atmospheric sciences (Rayleigh and Mie scattering) and signal theory (coherent detection and balanced detection), in order to understand and synthesise the design of a Doppler wind LiDAR system. I have also learnt how to use Python packages such as OptiCommPy for photonics simulation of systems, and how to conduct noise analyses in systems.

The initial problem was ill-defined and the decision to take the route of designing a benchtop-scale Doppler wind LiDAR system was not immediately obvious. It was only after learning more about the field and evaluating areas of research in it that this decision was made, after reaching out to my supervisor to consult on this. Throughout the project I have had to manage the scope of the project to suit my abilities and constraints, given that the field of research contains many complexities which are beyond an undergraduate skill-level and time-frame. This therefore shows my ability to synthesise meaningful objectives, knowledge and outcomes from an ill-posed problem.

Supervisor Response:
The student demonstrated strong independent learning through acquiring knowledge in several unfamiliar and multidisciplinary areas including: photonics, atmospheric sciences and signal theory. The student then applied acquired knowledge to define project objectives, evaluate design alternatives and manage project scope within practical constraints. The student is therefore on track to meet the requirements for GA-9.

GA 1: Problem Solving

Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.

- A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline, much of which is at the forefront of the discipline.

GA 4: Investigations, Experiments and Data Analysis

Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

The balance of investigation and experiment should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

Note: An investigation differs from a design in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

GA 5: Use of engineering tools

Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.

- Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- Knowledge of engineering practice (technology) in the practice areas in the engineering discipline

A range of techniques, resources and modern engineering and IT tools appropriate to the disciplinary designation of the programme.

GA 6: Professional and Technical Communication

Demonstrate competence to communicate effectively, both orally and in writing, with engineering audiences and the community at large, taking into account cultural, language, and learning differences.

This course evaluates the long report component of this outcome at exit level. Material to be communicated is in an academic or simulated professional context. Audiences range from engineering peers, management and lay persons, using appropriate academic or professional discourse. Written reports (10 000 to 15 000 words plus tables, diagrams and appendices) should cover material at exit-level. Methods of providing information include the conventional methods of the discipline, for example engineering drawings, as well as subject-specific methods.

GA 8: Individual, Team and Multidisciplinary Working

Demonstrate competence to work effectively as an individual, in teams and in multidisciplinary environments. This course evaluates the **individual** working component of this learning outcome at exit level.

Knowledge of professional ethics, responsibilities and norms of engineering practice.

GA 9: Independent Learning Ability

Demonstrate competence to engage in independent learning through well developed learning skills.

Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

Operate independently in complex, ill-defined contexts requiring personal responsibility and initiative, accurately self-evaluate and take responsibility for learning requirements; be aware of social and ethical implications of applying knowledge in particular contexts.

- Openness to constructive feedback, awareness of own limitations, ability to cope with the discomfort of uncertainty and having access to a range of approaches, reflective selfevaluation, curiosity and proactive engagement, resilience, confidence to ask for help and draw from a broad range of stakeholders.
- Reflection of self-learning to begin to recognise if what has been covered meets the needs of the activity or task.